



MEDIACREST
TRAINING COLLEGE

The Clinical Decision-Making and Policy Focus

Logistic Regression for Heart Disease Prediction

Victor Wandera Lumumba, MSc.

Statistician | Data Analyst | ML Expert

Mediacrest Training College

1st September 2026

Abstract

This presentation bridges the gap between statistical modeling and real-world clinical policy, utilizing a cardiovascular dataset to demonstrate decision-making frameworks.

- **Foundations:** Clinical EDA, stratified prevalence analysis, and robust missingness handling.
- **Modeling:** Building parsimonious logistic regression models tailored for risk stratification.
- **Translation:** Converting complex statistical outputs (like Odds Ratios) into intuitive marginal probabilities for non-technical stakeholders.
- **Policy & Thresholding:** Scenario-based threshold selection (Primary Care vs. Specialist Referral) balancing clinical trade-offs and financial costs.

We conclude with actionable recommendations for integrating predictive models into national Health Management Information Systems (HMIS).

Session Objectives

- Perform clinical EDA and stratified prevalence analysis to understand risk factor distributions.
- Apply clinically sound imputation strategies for missing data (e.g., median for ordinal, explicit categories for nominal).
- Formulate and fit a parsimonious logistic regression model for risk stratification.
- Translate model coefficients into actionable insights, avoiding the "Odds Ratio vs. Risk Ratio" trap.
- Evaluate model performance using ROC-AUC and Precision-Recall curves.
- Determine scenario-specific classification thresholds based on clinical goals (Maximizing Recall vs. Precision).
- Outline next steps for HMIS integration and algorithmic bias monitoring.

Part 1: Clinical EDA & Risk Factor Profiling

"Understanding the data before modeling the risk."

Target Audience, Dataset & Dependencies

- **Target Audience:** Analysts transitioning to rigorous statistical modeling and clinical decision support.
- **Dataset:** `Heart.csv` (Clinical cardiovascular data).
- **Dependencies:**
 - Data Manipulation: `pandas`, `numpy`
 - Statistical Modeling: `statsmodels`
 - Machine Learning: `scikit-learn`
 - Visualization: `matplotlib`, `seaborn`

1.1 Loading the Data

- Clinical datasets often contain inconsistent missing-value representations.
- **Best Practice:** Explicitly define missing value strings during data ingestion to prevent them from being treated as valid categories.

Python Implementation

```
1 import pandas as pd
2
3 # Treat common missing-value strings as missing
4 df = pd.read_csv(
5     'Heart.csv',
6     na_values=['NA', '?', 'NaN', ''])
7 )
```

1.2 Handling Missingness: Clinical Reality

Understanding Missing Data Mechanisms

- **MCAR:** Missingness is unrelated to any observed or unobserved data.
- **MAR:** Missingness is related to observed data (e.g., older patients might skip the Ca test).
- **MNAR:** Missingness is related to the unobserved value itself (e.g., patients with severe disease avoid the Thal scan).

Clinical Reality

In clinical data, MAR or MNAR is common. We must impute cautiously to avoid introducing bias or losing valuable “missingness” information.

1.2 Imputation Strategy

Variable	Type	Imputation Strategy
Ca	Ordinal (0–3)	Median imputation. Robust to outliers and preserves the ordinal nature of major vessel count.
Thal	Nominal	Create an explicit <code>unknown</code> category. This preserves the information that the test was missing, avoiding bias.

Best Practice

Avoid mean imputation for ordinal clinical scores. Preserving “missingness” as a distinct category in nominal variables prevents the model from making unfounded assumptions.

1.3 Feature Engineering: Age Bins

- Continuous variables like Age can be binned to improve clinical readability and facilitate stratified analysis.
- Implementation:** Create discrete, clinically meaningful age groups.

Python Implementation

```
1 df['Age_Bin'] = pd.cut(  
2     df['Age'],  
3     bins=[0, 50, 60, 70, 100],  
4     labels=['<50', '50-60', '61-70', '>70']  
5 )
```

Rationale: Allows us to analyze disease prevalence trends across distinct life stages rather than assuming a strictly linear relationship.

1.3 Feature Engineering: Sex Labels

- Numerical encodings (e.g., 0 and 1) are not intuitive for clinical reporting or stratified visualization.
- **Implementation:** Map numerical values to descriptive string labels.

Python Implementation

```
1 df['Sex_Label'] = df['Sex'].map({  
2     0: 'Female',  
3     1: 'Male'  
4 })
```

Rationale: Enhances the interpretability of downstream visualizations (e.g., countplots) and ensures clear communication with non-technical stakeholders.

1.4 Stratified Prevalence Analysis: Overall

- Before modeling, we must understand the baseline prevalence of the target variable (HD).
- **Observation:** This cohort is relatively balanced (approximately 50% prevalence).

Critical Note on Evaluation Metrics

Because the dataset is balanced, accuracy might seem like a good metric. However, clinical datasets are often imbalanced in the real world. We **must** evaluate metrics beyond accuracy, such as Precision, Recall, and ROC-AUC, to ensure robust clinical utility.

1.4 Stratified Prevalence: By Chest Pain

- Chest pain type is a primary presenting symptom in cardiovascular assessments.
- Stratifying prevalence by chest pain type reveals significant risk disparities.

Clinical Insight

Patients presenting with **asymptomatic** chest pain (or atypical presentations) often show the highest prevalence of underlying heart disease, highlighting the danger of relying solely on patient-reported symptoms.

(Visualization saved as `eda_chestpain_prevalence.png`)

1.4 Stratified Prevalence: By Age Bin and Sex

- Heart disease risk is known to compound with age and varies by biological sex.
- Cross-tabulating `Age_Bin` and `Sex_Label` allows for granular risk profiling.

Clinical Insight

The prevalence of heart disease steadily increases with age across both sexes. However, the rate of increase and baseline risk often diverge, necessitating sex-specific risk assessment protocols in older age brackets (>70).

(Visualization saved as `eda_age_sex_prevalence.png`)

1.4 Stratified Prevalence: By Sex

- Isolating the `Sex_Label` variable provides a high-level view of gender distribution in the cohort.
- This informs us about potential sampling biases or genuine demographic skews in the data collection process.

Monitoring for Bias

Understanding the baseline sex distribution is the first step in ensuring our final model does not perpetuate or amplify algorithmic bias against underrepresented groups during deployment.

(Visualization saved as `eda_sex_prevalence.png`)

Part 2: Logistic Regression for Risk Stratification

"Building a parsimonious model for clinical utility."

2.1 Defining the Parsimonious Clinical Risk Model

- A **parsimonious** model includes only the most predictive and clinically relevant variables, avoiding overfitting.
- We utilize `statsmodels` formula API for clear, R-style statistical specification.

Model Specification

```
HD ~ Age + Sex + C(ChestPain) + RestBP + Chol + Fbs + C(RestECG) + MaxHR + ExAng + Oldpeak + C(Slope)  
+ Ca + C(Thal)
```

Note: `C()` denotes categorical variables, with `statsmodels` automatically handling dummy encoding and reference levels.

2.1 Fitting the Logistic Regression Model

- **Method:** Maximum Likelihood Estimation (MLE).
- **Implementation:** `smf.logit(formula, data=df).fit(dis=0)`
- `dis=0` suppresses verbose convergence output for cleaner reporting.
- **Data Note:** Missing values were dropped (`df.dropna()`) prior to this specific model fit for demonstration, yielding a complete-case dataset.

Production Note

In a strict production pipeline, imputation should be fitted on the training set only to prevent data leakage.

2.1 Model Summary: Global Metrics

Metric	Value	Metric	Value
Dep. Variable	HD	No. Observations	297
Model	Logit	Df Residuals	278
Method	MLE	Pseudo R-squ.	0.5325
Log-Likelihood	-95.821	LLR p-value	2.124×10^{-36}

Interpretation

The LLR p-value (< 0.001) indicates the model fits the data significantly better than an intercept-only (null) model. A McFadden Pseudo R-squared of 0.5325 indicates a very strong fit for logistic regression.

2.1 Model Summary: Key Coefficients

Predictor	Coef	Std Err	z	P> z
Intercept	-6.0295	2.892	-2.085	0.037
ChestPain (asymptomatic)	2.0865	0.667	3.130	0.002
Sex	1.5460	0.530	2.917	0.004
RestBP	0.0244	0.011	2.162	0.031
Ca	1.3115	0.279	4.696	0.000

Note

Coefficients represent the change in the *log-odds* of heart disease for a one-unit change in the predictor, holding all other variables constant.

2.2 Translating Model Outputs for Stakeholders

- Statistical outputs (log-odds, p-values) are not intuitive for clinicians, patients, or policymakers.
- **Goal:** Translate model outputs into actionable, easily understandable metrics.

Translation Strategies

- 1 **Odds Ratios (OR):** Useful for researchers to understand relative effect sizes.
- 2 **Marginal Predicted Probabilities:** The gold standard for patient-facing risk communication (e.g., “You have a 25% estimated risk”).
- 3 **Absolute Risk Differences:** Clearly shows the impact of an intervention or risk factor.

2.2 The Prevalence Trap: OR vs. RR

Definitions

- **Odds Ratio (OR):** Native to Logistic Regression. Compares odds.
- **Risk Ratio (RR):** Compares probabilities (Risk). More intuitive for humans.

The Prevalence Trap

If the outcome is rare ($< 10\%$), $OR \approx RR$.

However, Heart Disease in this dataset is **~50% prevalent**. Therefore, the OR will **grossly exaggerate** the effect size compared to the RR.

Best Practice

Do not say “8 times more likely.” Explicitly state that OR is not RR, and use marginal predicted probabilities for patient-facing communication.

Part 3: Model Evaluation & Clinical Trade-offs

"Beyond accuracy: Evaluating real-world performance."

3.1 Train/Test Split Strategy

- **Purpose:** Avoid overly optimistic in-sample metrics and assess generalizability to unseen patients.
- **Implementation:**
 - `test_size=0.25` (75% train, 25% test)
 - `random_state=42` (Ensures reproducibility)
 - `stratify=y` (Maintains the proportion of Heart Disease cases in both splits)

Note on Demonstration

For the simplicity of visualizing curve shapes in this script, full dataset probabilities are used. In strict production, predictions must be mapped exclusively through `X_test`.

3.2 Fitting the Prediction Model

- The model is re-fit on the training data to simulate a real-world deployment pipeline.
- **Implementation:** `sm.Logit(y_train, sm.add_constant(X_train)).fit()`
- Predictions are then generated on the held-out `X_test` set.

Why re-fit?

Evaluating a model on the same data it was trained on leads to data leakage and inflated performance metrics. The test set must remain completely unseen during the training phase.

3.3 ROC-AUC Curve

Receiver Operating Characteristic

Plots True Positive Rate (Sensitivity) vs. False Positive Rate (1 - Specificity) across all thresholds.

Result:

ROC-AUC = 0.935

Interpretation:

Excellent ability to distinguish between patients with and without heart disease.

Visual Output

The curve (saved as `roc_curve.png`) bows sharply toward the top-left corner, far above the diagonal line of no-discrimination ($AUC = 0.5$).

3.3 Precision-Recall Curve: Clinical Trade-offs

Why PR Curve?

Crucial for high-stakes clinical data where False Positives and False Negatives carry heavy, asymmetric costs.

Result:

PR-AUC = 0.929

Interpretation:

Demonstrates strong predictive performance in balancing Precision (Positive Predictive Value) and Recall (Sensitivity).

Clinical Relevance

In heart disease screening, a high Precision-Recall AUC ensures that when the model flags a patient as high-risk, it is highly likely to be a true positive, minimizing unnecessary, invasive follow-up procedures.

Part 4: Threshold Selection for Clinical Policy

"One size does not fit all: Scenario-based decision making."

4.1 The Challenge of Default Thresholds

- Machine learning libraries default to a classification threshold of **0.5**.
- **The Problem:** A 0.5 threshold assumes that the cost of a False Positive (unnecessary test) is exactly equal to the cost of a False Negative (missed diagnosis).
- **Clinical Reality:** These costs are rarely equal. The optimal threshold depends entirely on the **clinical context** and **resource availability**.

Policy Directive

Thresholds must be explicitly chosen based on the specific healthcare scenario, not left to software defaults.

4.2 Scenario A: National Screening Program (Primary Care)

The Clinical Goal

Maximize Recall (Sensitivity). We want to catch as many true cases of heart disease as possible in the general population.

- **Trade-off:** We accept more False Positives.
- **Rationale:** In primary care, it is cheaper and safer to re-test a healthy patient (False Positive) than to miss a life-threatening condition (False Negative).

Threshold Strategy

Shift the threshold **lower** (e.g., < 0.5) to cast a wider net, ensuring high sensitivity.

4.2 Scenario A: Maximizing Recall (Sensitivity)

- By lowering the threshold, the model flags more patients as “high risk.”
- **Benefit:** Very few true heart disease cases are missed (Low False Negative Rate).
- **Cost:** Increased burden on the healthcare system due to unnecessary follow-up tests for healthy individuals (High False Positive Rate).

Implementation

This strategy is ideal for Community Health Promoters (CHPs) or mobile screening clinics where the primary objective is triage and referral, not definitive diagnosis.

4.3 Scenario B: Specialist Referral (Cardiology Clinic)

The Clinical Goal

Maximize Precision. We want to ensure that patients referred for invasive, expensive procedures (like angiograms) truly have the disease.

- **Trade-off:** We accept more False Negatives.
- **Rationale:** In a specialist setting, the cost (financial and physical) of an unnecessary invasive procedure is very high. We must be highly confident before acting.

Threshold Strategy

Shift the threshold **higher** (e.g., > 0.5) to ensure only the highest-risk patients are flagged.

4.3 Scenario B: Maximizing Precision

- **Target Precision:** 0.85 (We want 85% of flagged patients to truly have the disease).

Metric	Value
Recommended Probability Threshold	0.658
Expected Precision	85.3%
Expected Recall (Sensitivity)	82.9%

Interpretation

At a threshold of 0.658, the model achieves the desired 85.3% precision, ensuring that specialist resources are allocated efficiently, while still maintaining a respectable recall of 82.9%.

4.4 Cost-Benefit Analysis of Thresholds

- Threshold selection is fundamentally an economic and ethical optimization problem.
- **Example Cost:** A False Positive Angiogram may cost approximately **20,000 Ksh**, plus patient anxiety and procedural risk.
- **Example Cost:** A False Negative may result in an unmanaged myocardial infarction, with catastrophic health and financial outcomes.

Policy Directive

Healthcare administrators must explicitly define the acceptable cost ratio of False Positives to False Negatives. The model threshold should then be mathematically optimized to minimize the *expected total cost*.

Part 5: Deliverable & Recommendations

"From code to clinical impact."

5.1 Clinical Decision Support Report: Data & Model

1. Data Preparation

- **Missingness:** Ca imputed with median (0.0); Tha1 imputed as unknown.
- **Encoding:** Dummy variables applied to nominal features; ordinal features retained.

2. Model Fit & Odds Ratios

- **Pseudo R-squared (McFadden):** 0.533 (Substantial improvement over null model).
- **Top Predictors:** Asymptomatic Chest Pain (OR=8.06), Sex (OR=4.69), Reversible Thal (OR=4.03).

5.2 Clinical Decision Support Report: Diagnostics & Metrics

3. Diagnostics

- **Multicollinearity:** Max VIF = 1.35 (Age). No severe multicollinearity detected.
- **Linearity of Logit:** Visually inspected and holds.
- **Interaction Check:** ExAng * Sex p-value = 0.238 (Not significant; no effect modification).

4. Evaluation Metrics

- **ROC-AUC:** 0.935 | **Precision-Recall AUC:** 0.929
- **Recommended Policy Thresholds:** 0.481 (Balanced), 0.658 (High Precision).

6. Next Steps & Recommendations (Deployment)

Actionable Directives for Health Systems

- 1 **HMIS Integration:** Integrate this parsimonious model into the national Health Management Information System (HMIS) or develop a lightweight mobile tool for Community Health Promoters (CHPs).
- 2 **Prospective Pilot:** Conduct a pilot study in 2–3 sub-county hospitals to validate these scenario-specific thresholds on local Kenyan demographic data before nationwide rollout.

6. Next Steps & Recommendations (Governance)

Actionable Directives for Health Systems (Cont.)

- ③ **Bias Monitoring:** Ensure continuous monitoring for algorithmic bias across gender and age groups. Disparate impact must be actively measured and mitigated.
- ④ **Model Retraining:** Establish a pipeline for periodic model retraining as new clinical data becomes available, ensuring the model adapts to shifting population health trends.

Overall Conclusion

- The diagnostic assessment indicates the logistic regression model has **good statistical properties** and **excellent predictive performance** (ROC-AUC = 0.935).
- There was no evidence of problematic multicollinearity, and model assumptions were reasonably met.
- Asymptomatic chest pain, Sex, and reversible Thal were among the strongest, statistically significant predictors.
- **Crucial Takeaway:** Clinical interpretation must emphasize **marginal predicted probabilities** rather than Odds Ratios alone, due to the high prevalence of the outcome in this cohort.
- Thresholds must be dynamically adjusted based on the specific clinical scenario (Primary Care vs. Specialist Referral).

Questions?

Victor Wandera Lumumba

Statistician / Data Analyst / ML Expert

Mediacrest Training College

Nairobi, Kenya

Email: [lumumbavictor172@gmail.com](mailto:lumbavictor172@gmail.com)

Web: beyonddataanalytics.online

GitHub: [Lumumba1992](https://github.com/Lumumba1992)